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PROJECT REPORT

MCDA5585 and MCDA5586

Taken in Lieu of:

MCDA 5550 & 5510

Submitted By:

Mr. Evgeny Chudaeu

A00280004

Under the guidance of:

Dr. Pawan Lingras

Mrs. Cyndy Allen

In partial fulfillment for the awarding of the degree:

Master of Science in

Computing and Data Analytics

October 2023

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CERTIFICATE

This is to certify that Mr. Evgeny Chudaev (ID: A00280004), a student at Saint Mary's University in the Master of Science in Computing and Data Analytics Program, has completed the project in fulfillment of the course requirements.

.....
Dr. Pawan Lingras

ACADEMIC SUPERVISOR

DEPARTMENT OF MATHEMATICS

AND COMPUTING SCIENCE

SAINT MARY'S UNIVERSITY

923 Robie St. Halifax, NS, Canada B3H 3C3

.....
Mrs. Cyndy Allen

DIRECTOR, TOTAL REWARDS, HR

EMERA INC.

5151 Terminal Rd., Halifax, NS, Canada

ACKNOWLEDGEMENT

I want to express my gratitude to all the faculty and staff of the Master of Science in Computing and Data Analytics program at Saint Mary's University and Dr. Pawan Lingras. Your commitment to student support and excellence in education has enabled MCDA students like me to accomplish ambitious educational, professional, and personal goals during this program.

This report is also a result of a professional commitment to my employer, team, and colleagues at Emera Inc., who meaningfully supported me throughout this graduate program in various ways. I am grateful for your support, allowing me to work for a fantastic organization and accommodating my educational needs over the last year and a half.

Lastly, I thank my family for supporting my educational aspirations. You have helped me balance my studies with family life and motivated me to reach new levels.

ABSTRACT

This report will introduce the reader to the project I have completed to predict voluntary employee attrition in Canadian affiliates of the Emera group of companies which includes over 8 different affiliates. The project has resulted in the creation of the first predictive model related to workforce attrition at Emera and several artifacts such as research notes, code repository, graphical user interfaces, and a strong foundation for subsequent research in this domain as well as deployment of the model to production in Azure cloud later.

Despite the management role, I was the sole developer and a data scientist on this project due to time constraints faced by my team and a requirement to complete this project to satisfy the requirements of the MCDA program. I have learned a great deal about the practical application of Machine Learning for solving business problems as well as the strengths and limitations of different Machine Learning algorithms for different business problems.

Finally, as this model may be deployed to production at some point, this project has also enabled me to practice writing better code that follows some best practices such as naming conventions, readability, scalability, efficiency, and utilization of appropriate data structures.

PROJECT BACKGROUND

Company Background

Emera is one of Atlantic Canada's largest publicly traded companies and a North American Top 20 utility. From its origins as a single electric utility in Nova Scotia, Emera has grown into an energy leader with \$40 billion in assets and serves 2.5 million customers in Canada, the US, and the Caribbean.

Emera's strategic focus for more than 15 years has been to safely deliver our customers cleaner, affordable, and reliable energy. Emera is focused on making investments in cleaner and renewable sources of energy, modernizing our infrastructure and customer-focused technologies while maintaining affordability, enhancing reliability, adopting emerging technologies, and working constructively with policymakers, regulators, partners, investors, and communities.

With a portfolio of high-quality utilities and an innovative approach to developing solutions for current and future energy needs, Emera is well-positioned to continue delivering for customers while also providing long-term growth in earnings, cash flow, and dividends to our shareholders.¹

Team Composition

The HR process and analytics team is based in Halifax and comprises a manager and HR data analyst. The HR data analyst is a direct report of an HR analytics manager who, in turn, reports to a senior manager of the Total Rewards team.

¹ 2022 Emera annual report. Retrieved on 7-July-2023 from [Emera Inc. 2022 Annual Report \(q4cdn.com\)](https://www.emera.ca/en/2022-annual-report)

The HR analytics team has numerous responsibilities concerning analytics and process improvement functions, such as providing timely business intelligence and analytics to senior management and executive members of Emera, providing thought leadership in People and HR process analytics, and supporting all Emera affiliates with expert advice and sharing best practices on analytics, reporting and data governance; finding and promoting initiatives to realize process efficiencies and improving HR and payroll processes using data analytics and evidence-based decision making.

To that end, the team collaborates with various stakeholders inside and outside the organization and analysts from other affiliates who make up a larger analytics function at the organization.

The core HR process and analytics manager and HR data analyst team work together at the parent company of Emera – Emera Inc. The professional backgrounds and team members' experience include industrial engineering, computer science, business intelligence, and analytics. Both core HR process and analytics team members have significant experience with various tools, techniques, programming languages, and frameworks used in data science and business intelligence. Both team members often work together on projects impacting the entire organization and its affiliates and on one-time custom ad-hoc data and information requests from business representatives, thereby supporting and promoting evidence-based decision-making at Emera.

My Role

Despite the management role, I was the sole developer and a data scientist on this project due to time constraints faced by my team and a requirement to complete this project to satisfy the requirements of the MCDA program. Nevertheless, my team has taken on additional

responsibilities and some daily tasks off my plate while I was completing this project to free up my time so I could concentrate my attention on this project.

As an individual contributor to this project, I have learned a great deal about the practical application of Machine Learning for solving business problems as well as the strengths and limitations of different Machine Learning algorithms for different business problems.

More specifically, my accountabilities in this project included the following:

- Determining business objectives.
- Collecting business and functional requirements.
- Surveying the related literature.
- Deep dive into various data science frameworks such as CRISP-DM.
- Data cleansing, preparation, and profiling.
- Performing an exploratory data analysis (EDA).
- Data pre-processing and feature engineering for modeling.
- Model selection, training, fine-tuning, and evaluation.
- Model serialization and deployment.
- Developing a graphical user interface for the end users of the model.
- Model interpretation and presentation to senior leaders.

PROJECT OVERVIEW

Project History

Emera HR has been doing Employee attrition analytics and reporting on an ongoing basis for many years as part of quarterly reporting and various regulatory filings. However, these initiatives have either been providing limited insights or have not looked into the future of workforce attrition to predict potential risks for business continuity. The need for predictive modeling for employee attrition has become especially evident right after the pandemic when many organizations have faced a pent-up turnover and the prospect of a 'great resignation.'

Having joined Emera Inc. Corporate HR unit in 2022, I have immediately identified the need for more consistent, reliable, and in-depth reporting and analytics related to turnover and my team has started working on several reporting initiatives to further this initiative. We built several Tableau dashboards dedicated to workforce attrition at first and this was a huge step forward for us and our organization.

Nevertheless, as good as the descriptive analysis was, the question of who exactly is likely to resign, when, and why could not be answered. This has motivated me to work on inferential and predictive modeling related to the topic of employee attrition. I got very excited about this initiative and at first, surveyed the related literature and looked for similar projects on GitHub. To my surprise there were very few comprehensive projects and articles on this subject matter and those that were available did not seem like they were written by the practitioners. This has not discouraged me at all, and I decided to bring on the challenge despite the uncertainties.

Project Motivation

Employee attrition, the process of employees leaving an organization, is a critical concern for businesses of all sizes and industries and Emera is no exception. High employee turnover can have significant negative impacts on our organization, including increased recruitment costs, decreased productivity, loss of institutional knowledge, and reduced employee morale.

Therefore, it is essential for Emera to proactively manage and reduce employee attrition. One effective approach to addressing this challenge is the implementation of predictive modeling techniques which can help achieve several objectives such as:

- **Identifying Key Contributors to Attrition:** Predictive modeling will allow us to identify the key factors that contribute to employee attrition. By analyzing historical data, Emera can determine which variables, such as job satisfaction, salary, workload, or tenure, are strongly correlated with attrition. This insight will enable senior management to address the root causes of attrition more effectively, whether through changes in policies, compensation adjustments, or targeted employee engagement initiatives.
- **Early Warning System:** Predictive models can serve as early warning systems for potential attrition. By continuously monitoring employee data, these models can detect patterns or anomalies that suggest an employee may be at risk of leaving the organization. This may allow Emera leadership to intervene proactively, engaging with employees to address concerns, provide support, and increase the chances of retaining valuable talent.
- **Cost Reduction and Resource Optimization:** Employee turnover is expensive. Recruitment, onboarding, and training of new employees incur substantial costs. Predictive modeling can help us reduce these expenses by decreasing the frequency of attrition. As a result, Emera can allocate its resources more efficiently by focusing retention efforts on employees identified as high-risk by the predictive model.
- **Talent Retention and Organizational Stability:** High attrition rates can lead to a lack of stability within an organization, impacting team dynamics, project continuity, and overall performance. Predictive modeling may help us maintain a stable workforce, ensuring that the skills and knowledge of experienced employees are retained. This will undoubtedly contribute to improved productivity and overall organizational success.
- **Competitive Advantage:** In today's competitive job market, attracting and retaining top talent is a key differentiator. Organizations that effectively use predictive modeling for employee attrition gain a competitive advantage by creating a work environment that

fosters employee satisfaction and loyalty. This, in turn, enhances the organization's reputation, making it an attractive destination for skilled professionals.

Thus, predictive modeling for employee attrition is not just a valuable tool; it's a strategic imperative for Emera. By leveraging data-driven insights to proactively address attrition, Emera can enhance its talent retention efforts, reduce costs, and maintain a competitive edge in the job market. It is a wise investment that can lead to a more stable and successful organization in the long run.

Project Scope

I have decided on the scope of the project after consultation with the senior leadership team, doing research on similar initiatives, and considering my constraints such as time and other priorities. To make this project delivery feasible and manageable, I have broken it into several incremental phases:

- Phase I – Exploratory analysis (EDA), data cleansing, preparation, and feature engineering.
- Phase II – Modeling, model training, model tuning, and model selection.
- Phase III – Model interpretation, insights generation, and deliverable presentation.
- Phase IV – Model deployment to production.

So far, I have been able to complete the first 2 phases and left out the remaining 2 phases for the upcoming winter. The model deployment is subject to the availability of infrastructure and is optional as the primary objective of this project is to generate insights and provide recommendations to the senior leadership at Emera. However, to showcase some tangible results, I have been able to create a tool with a simple graphical user interface that can be used by the end users who want to predict the attrition for an employee or a group of employees.

PROJECT GOALS

As discussed in the previous section, the project has several phases and only the first two phases were pursued at this time with the remaining ones to be completed in the upcoming winter or later.

Thus, the relevant goals include the following:

- Creating a pipeline for data cleansing, preparation, and profiling.
- Performing all pre-processing and feature engineering steps for modeling.
- Performing an explanatory data analysis to get business intuition for predictive models.
- Training, fine-tuning, evaluating, and ranking models by predictive power.
- Generating initial insights from each model to be used for the executive report in the future.

LEARNING GOALS

The learning goals for this project include the following:

Table 1: List of Learning Goals

No	Goals	Tasks
1	Practice applying Machine Learning algorithms to solve business problems	Learn how to translate high-level business requirements into a detailed set of subproblems that can be addressed by Machine Learning.
		Create a simple Machine Learning pipeline including activities such as data assimilation, transformation, preprocessing, modeling, and prediction.
2	Practice writing maintainable code	Practice using naming conventions and best practices
		Emphasize code readability, maintainability, and scalability.
3	Practice application of CRISP-DM framework for data mining	Practice using a disciplined approach for data mining project activities.

4	Learn several new Python packages	Learn several new libraries and practice using well-known libraries such as Scikit-learn, Keras, XGBoost, and Tkinter.
5	Practice model tuning and model selection	Practice using model tuning techniques such as grid search, Bayesian optimization, random search, etc.

ASSUMPTIONS

There were several assumptions that I have made about the workforce, environment, end users, and technology:

- The workforce's propensity for turnover is explained by various demographic, professional, and personal characteristics and does not change over time.
- The environment has a limited impact on employees' decision to leave the organization. While it is a significant simplification, it would not be possible to account for and predict events such as pandemics, etc.
- End users would prefer the model to explain drivers of workforce attrition in detail rather than predict exactly when each employee is likely to resign.
- The available infrastructure currently does not allow for model deployment and model training can only be done on my local machine at this time.
- Azure cloud infrastructure may be available in the future for deployment and model tuning, subject to negotiation with IT.

REQUIREMENTS GATHERING

I have gained an understanding of the business objectives related to workforce planning based on my consultation with the senior leadership at Emera and have turned those objectives into concrete functional requirements using my vision of the final product and the necessary deliverables that would inform the senior leadership team at my organization.

Therefore, I have come up with the following list of requirements that would produce meaningful insights for stakeholders and will eventually lead to a consumable product used at Emera:

Model Requirements

The model should produce a reasonably accurate prediction for employees' resignation. Since the outcome variable that we are trying to predict is "whether or not an employee will resign within 4 years of the hiring date", it is a binary classification problem. The most suitable performance metric is the ROC AUC score and our objective was to minimize false negative rate even at the expense of increased false positive rate. The minimum ROC AUC score for the model to be considered for future use is 70%. Several models should be trained, tested, tuned, and evaluated for performance and accuracy so that we can be sure about the strengths and limitations of each one.

Business Requirements

No matter how fancy the model may be, the only thing senior leaders at Emera would be interested in are factors affecting employees' decisions and the probability of resigning. Hence, the model should explain the most important factors and their contribution to the model. These and other insights should be clearly explained in non-technical language in the report.

Product Requirements

The ultimate goal of the project is to deploy the model to the production environment and design a simple front-end web interface where end users such as functional managers can predict resignations within their teams and act proactively to address the business continuity problems. This requirement will be satisfied in the future when the necessary infrastructure is provided by IT.

Software Requirements

Below is a list of software and tools used in this project.

- Python libraries: Scikit-learn, Keras, Pandas, Matplotlib, NumPy
- Tableau
- Microsoft Power BI
- Excel

Hardware Requirements

Below is a list of hardware requirements.

High-performance laptop with the following specifications:

- CPU: 11th Gen Intel(R) Core(TM) i7-11800H @ 2.30GHz 2.30 GHz
- RAM: 32 GB
- System type: 64-bit operating system, x64-based processor

PROJECT DEVELOPMENT METHODOLOGY

The project development methodology for this project can be loosely called Agile (Scrum variant) without me strictly following all the procedures. While I followed many principles native to the family of Agile methodologies (particularly Scrum), such as backlog management, sprint reviews, and retrospect, I've also understood that as a sole contributor, there is no one else who can participate in daily stand-up meetings so this and other Agile ceremonies were unnecessary.

LITERATURE SURVEY

Keith McNulty in his book *"Handbook of Regression Modeling in People Analytics"* sheds light on practical applications of regression models in the People Analytics domain. This book was essential for my understanding of not only model inner workings but also how to interpret outputs and coefficients of regression models and assess their suitability. The author emphasizes the need for careful model interpretation and evaluation considering the typically smaller datasets used in the People Analytics domain.

Another great book referenced in this project that explains the most common Machine Learning frameworks and algorithms is the *“Hundred Page Machine Learning Book”* by *Andriy Burkov*. Despite its telling name, the book covers all the essentials and helps you build an intuition behind these models.

The *“Practical Statistics for Data Scientists”* by *Peter Bruce, Andrew Bruce, and Peter Gedeck* provided me with a phenomenal overview of Machine Learning Libraries in R and Python. I have encountered several practical problems and techniques described in the book that were later used in the project. One of the most important lessons I have learned from this book is how to deal with multicollinearity in the data and why this phenomenon can lead to spurious results if not considered.

The last book referenced in this project is *“Essential Math for Data Science”* by *Thomas Nield*. The author builds the mathematical intuition behind machine learning algorithms necessary for any data scientist. The book also provides a glimpse into the world of probability and statistics as well as computational methods used in several key Python libraries such as NumPy and SciPy.

IMPLEMENTATION

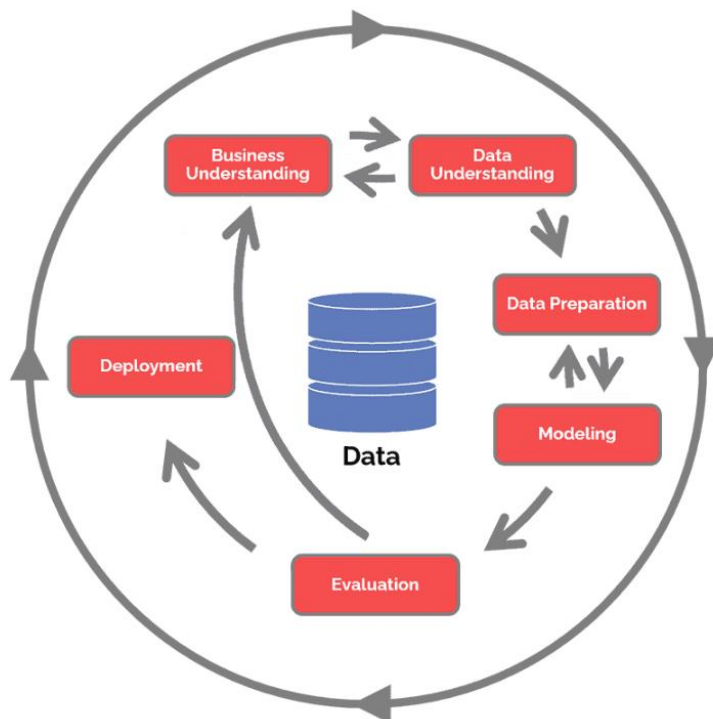
As mentioned earlier, in this project, I have followed the CRISP-DM framework that standardizes the data mining process across different domains and industries and has become the most common methodology in data science, analytics, and related projects.

The framework is depicted in the below diagram and comprises the following phases:

- Business understanding – understand business needs and objectives.
- Data understanding – understand what data is available and what transformations on data are needed.
- Data preparation – prepare and process data for modeling.

- Modeling – understand what models/algorithms can be used to solve business problems.
- Evaluation – decide what model/algorithm best suits the business objectives.
- Deployment – decide how to deploy the model/algorithm and provide users with access to results.

Figure 1: CRISP-DM Diagram. Source: What is CRISP-DM? - Data Science Process Alliance (datascience-pm.com)



I will describe each phase and activities I have undertaken within the respective phase in detail below.

1. Business Understanding

In this initial phase, I have worked closely with stakeholders to understand the business problem and objectives. I had been doing this phase for a long time before the project began.

I've defined what a desired outcome of the project looks like and how data science can address the problem.

In the business understanding phase, I have identified key business metrics (e.g., percentage of employee resignations that can be predicted) and learned how reducing turnover would impact our business. I have documented business understanding and assumptions in a plain text file that I used for reference throughout the project. In this phase, I have laid the foundation for the entire project, aligning data science efforts with business objectives.

2. Data Understanding

After I achieved a clear understanding of the business problem, my focus shifted to the data. In this phase, I have collected and explored the available data sources to ascertain their relevance and quality for the project. The sole source of the data for this project was the PeopleSoft Human Resource Information System at Emera containing employee master data and all transactions for each employee including promotions, hiring details, vacation as well as sick leave data, compensation, and many more.

I have examined the data's structure, identified potential data gaps, and assessed data quality issues that may need to be addressed in the data preparation phase. My goal was to ensure that the data was comprehensive and suitable for analysis, paving the way for data preparation and modeling phases.

3. Data Preparation

In this phase, I focused on getting the data into a suitable format for analysis, addressed issues, and prepared data for modeling. This phase was essential to ensure that the data used in subsequent steps was of high quality and relevant to the project's objectives.

Firstly, I have performed data collection and integration during this phase. I have gathered data from various sources within the HR Information system database and consolidated data into a single dataset using core Python and various packages such as Pandas and NumPy. I have built a

simple yet efficient data integration pipeline that was easy to update if new sources were to be incorporated.

For example, in this project, I have engineered a dataset consisting of employee demographics, salary history, performance evaluations, and historical attrition records. Integration involved harmonizing data formats and units, and ensuring that the data from different sources can be effectively combined.

Secondly, I have emphasized data cleaning and preprocessing in this phase. I have identified and handled missing values, outliers, and inconsistencies in the dataset. In the context of this project, I've dealt with missing data in the form of incomplete employee records or addressing outliers that may skew the analysis. I have also standardized and scaled variables to ensure that they have a similar range and magnitude. I have used a variety of techniques and Python libraries in this activity such as min-max scaler, NumPy, Pandas, and visualization libraries such as matplotlib. Occasionally, I have also used MS Power BI and Tableau for quick and interactive analysis of possible outliers.

Lastly, I have performed feature engineering and feature selection in this phase. Feature engineering involves creating new variables or transforming existing ones to capture meaningful information that can improve model performance. This part has taken me a significant amount of time and effort not because of poor data quality but because the logic of many of the variables derived from historical data that were later used in the model was rather complex and had many nuances that were not immediately clear and was dependent on time, employee class and other factors. I have checked each variable and feature generated in this phase thoroughly against the raw data to make sure correct business logic was used to derive it. Overall, it was a very grinding and iterative process with a lot of rework and revision of assumptions about the data.

Overall, in this phase I have created of a dataset represented by several dozen features such as "average time between promotions", "age at hire", "annual compensation increase" etc. Feature selection using business knowledge, common sense, and trial and error helped me identify the most relevant variables for modeling, helping to reduce dimensionality and

enhance model interpretability. I have not had to use dimensionality reduction techniques such as PCA due to the limited number of features compared to some other projects in other domains.

4. Modeling

The Modeling phase is a pivotal step in developing predictive models to address specific business objectives. In my project, I have focused on selecting appropriate algorithms, and training models, and optimizing them for accurate predictions.

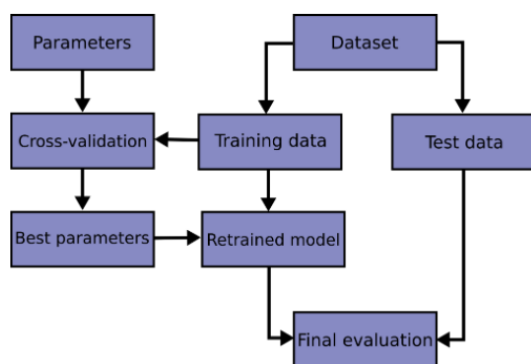
First, I have evaluated various algorithms suitable for the problem at hand. For instance, I have used logistic regression, decision trees, random forests, gradient-boosted trees, SVM, and several other methods. I deliberately did not use any deep learning methods as the key objective of this project is the interpretability of the model and deep learning algorithms are rather challenging to interpret and explain to decision-makers.

Secondly, I have split the dataset into training and testing sets to train and evaluate the model's performance. Then, I used the training set for 5-fold cross-validation and selection of best hyperparameters and subsequent retraining. I have fed historical data on employee attributes, into the selected algorithms. The model learned patterns from this data to make predictions. In the case of employee resignation, my goal was to predict which employees are likely to resign based on historical patterns. I have used the k-fold cross-validation technique in this project given a limited dataset size whereas a simpler hold-out cross-validation technique would likely have produced a less stable and less trustworthy model. The schematic explanation of the cross-validation, retraining, and testing is described in the diagram adopted from the scikit-learn documentation below.

Lastly, I have conducted hyperparameter tuning and model evaluation to optimize the model's performance. I have tried several techniques to optimize model parameters such as grid search, random search, and Bayesian optimization technique. Evaluation metrics included ROC AUC score, accuracy, precision, recall, and F1-score. The goal of this activity was to ensure that the model is not only accurate but also effectively identifies employees at risk of resignation. Given

the limited compute power I have only attempted model optimization on a limited grid space. While doing so has improved model performance, I have identified additional opportunities for further improvement with a wider variety of hyperparameters and extended grid space to be considered in the future.

Figure 2: Cross-validation – evaluating estimator performance. Source: 3.1. Cross-validation: evaluating estimator performance — scikit-learn 1.3.1 documentation



5. Evaluation

As mentioned above, I have performed the evaluation of estimators within each model (model instances) using metrics such as ROC AUC score and other metrics derived from the confusion matrix on cross-validation and test sets to ensure the best predictive power for each type of model. I have then compared individual models with each other based on the same set of metrics and ranked them according to performance as follows (from best to worst):

- XGBoost
- Random Forest
- Logistic Regression
- Decision Tree
- SVM

Finally, I have logged performance metrics of each model and used for the above ranking and future analysis. It is important to log various model parameters and distributions as even the

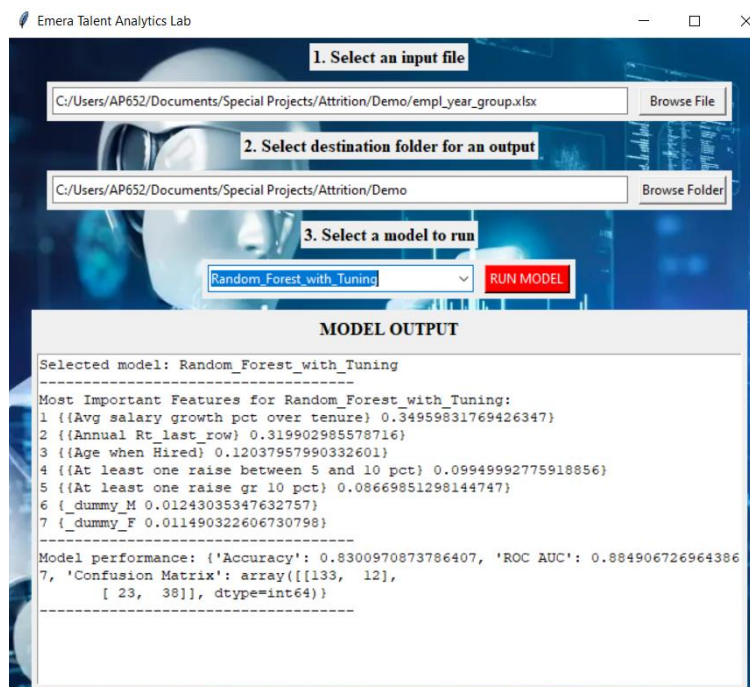
best models may require periodic retraining to avoid data drift issues (i.e. when data distributions change over time).

6. Deployment

In my project, the deployment phase involved serializing the model (i.e. saving the model state to a binary file) for future retrieval and for making predictions. Since I did not have a needed infrastructure available for hosting the model endpoint, the model deployment was done as follows:

- I serialized the model using the Python 'pickle' package.
- I have loaded serialized model from the 'pickle' file into the classifier object on my laptop at the time when prediction needs to be done.
- I have created a simple graphical user interface using the Python package 'tkinter' to perform model loading, data file upload, model predictions, and saving of output of predictions to a file. The graphical interface is shown in the screenshot below.

Figure 3: A graphical user interface for the desktop app for employee attrition predictions.



RESULTS

I have achieved the following results in this project:

- A working set of predictive models tuned and ranked with each other in terms of performance.
- Ensemble models such as Random Forest and XGBoost have vastly outperformed more simple models like logistic regression, decision trees, and SVM in this problem domain. The best-performing model had a ROC AUC score of close to 80%.
- K-fold cross-validation has dramatically improved model performance and generalization on unseen data while minimizing model overfitting.
- Feature importance analysis output by the model has provided me with insights for an executive report to be completed in the future.
- Dimensionality reduction techniques such as PCA were unnecessary due to the small number of features used in the analysis. The number of features was already reduced after less influential features were dropped.
- A simple graphical user interface was designed for users to make predictions on demand from their laptops while we were waiting for a cloud infrastructure from IT.

ACHIEVEMENTS

There were several notable achievements as a consequence of this project:

- I have launched the first successful iteration of the predictive model that has high performance and can be generalized to Emera affiliates.
- I have laid the foundation for future projects in the Machine Learning space in HR by showing leadership the power of AI in solving business problems.
- I have developed an automated end-to-end pipeline for data assimilation, preparation, model training, and predictions.

- I wrote a well-documented code following industry best practices for maintainability and scalability.
- I explored ways to deploy models to production in Azure cloud and began discussions with IT on obtaining the necessary infrastructure in the future.

All in all, my employment at Emera Inc. will continue for the foreseeable future, and I will be committed to pushing our organization further up the data analytics maturity model by providing insights using Machine Learning techniques in addition to descriptive analysis.

CHALLENGES

One of the main challenges involved a lack of infrastructure and limited computing power. I had to do all the work from a local machine which resulted in model training taking a significant amount of time, especially for ensemble models like random forest with hundreds of trees. Had the training dataset been larger, it would have been nearly impossible to train more complex models within a reasonable amount of time.

Another notable challenge relates to the quality of data. There were many inconsistencies found in the data due to potential business process problems and hence some variables could not be used in the model.

The last challenge was unrelated to the project itself and involved time constraints on my side as I had to balance multiple priorities at work besides working on this project. As a manager, I have a supervisory responsibility in addition to other ongoing work that requires my time. Nevertheless, I was able to complete this project thanks to the support of my team and my colleagues who took on additional responsibilities and supported me in this endeavor.

PROJECT GOALS REVIEW

Table 2: Outline of the status of project goals.

No	Goal	Status
1	Creating a pipeline for data cleansing, preparation, and profiling.	Completed
2	Perform all pre-processing and feature engineering steps for modeling.	Completed
3	Perform an explanatory data analysis to get business intuition for predictive models.	Completed
4	Train, fine-tune, evaluate, and rank models by predictive power.	Completed
5	Generate initial insights from each model to be used for the executive report in the future.	Completed

LEARNING GOALS REVIEW

Table 3: Outline of the status of learnings goals.

No	Goal	Status
1	Practice applying Machine Learning algorithms for solving business problems.	Completed
2	Practice writing maintainable code.	Completed
3	Practice application of CRISP-DM framework for data mining.	Completed
4	Learn several new Python packages.	Completed
5	Practice model tuning and model selection.	Completed

CONCLUSION

Summary

Overall, I have successfully completed the development of Emera's first predictive model related to workforce attrition. Additionally, I have produced several artifacts, including research notes, a code repository, graphical user interfaces, and a solid foundation for future research in this field. Furthermore, we plan to deploy the model to production in the Azure cloud at a later stage.

Through this project, I have gained valuable insights into the practical application of Machine Learning for addressing real-world business challenges. I have also gained a deeper understanding of the strengths and limitations of various Machine Learning algorithms when applied to different business problems.

Moreover, as there is a potential for this model to be deployed in a production environment in the future, this project has allowed me to enhance my coding skills. I have focused on adhering to best practices, including maintaining consistent naming conventions, improving code readability, ensuring scalability, enhancing efficiency, and utilizing appropriate data structures.

Hours Spent

Table 4: Outline of hours spent.

Project Phase	Hours
Business and data understanding	19
Data preparation	76
Modeling	57
Evaluation	48
Deployment	24
Documentation writing	20
Total Hours	244

Future Work

As a full-time employee of Emera Inc., I plan to continue my professional journey with this organization for the foreseeable future. I will continue impacting HR through analytics and automation. As I wrap up this project, I plan to start several other projects in the predictive analytics space and utilize Machine Learning algorithms to solve other complex business problems.

I also hope that my dedication, work ethic, and leadership skills will be noticed and considered for future career growth opportunities within the organization.

GLOSSARY OF TERMS

There are many acronyms and terms in the HR and BI/analytics world that may confuse someone unfamiliar with this subject area. Below is the definition of some of them used in this report.

Python – a high-level programming language often used by data analysts.

Tableau – is one of the leading business intelligence and visualization tools.

Power BI – another leading business intelligence and visualization tool.

BI – business intelligence. A process/function of extracting insights from the data.

SME – subject matter expert.

API – application programming interface.

HRIS – HR Information System. A system where HR data is stored, entered, and retrieved. It may also be linked to the applicant tracking system and Payroll module.

EDA – exploratory data analysis.

SCIKIT-LEARN – a popular machine-learning library for Python.

KERAS – another popular machine-learning library for Python.

TKINTER – a popular package for writing graphical user interfaces in Python.

PANDAS – data manipulation library in Python.

NUMPY – numerical computing library in Python.